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1- OVERVIEW

Components in the Magnet Enclosure Power Supply (MEPS) module are sent calibrated and adjusted from the factory. Procedures are provided here for checking and adjusting voltages because special circumstances—such as cable length—affect voltages to the boards.

- Check the voltages to the PAC, SRI, Patient Alignment Light, Bore Light, and Longitudinal/Dock power supplies from test points on the back of the MEPS module. This allows you to check the voltages without sliding out or opening the module.
- *If adjustments are necessary*, slide out, open the module, and make the necessary adjustments with test points on the boards.
- The PAC and SRI Power Supplies are located on the Low Current Power Supply Board.
- The Patient Alignment Light, Bore Light, and Longitudinal/Dock power supplies are located on the High Current Power Supply Board.

2- TEST EQUIPMENT REQUIRED

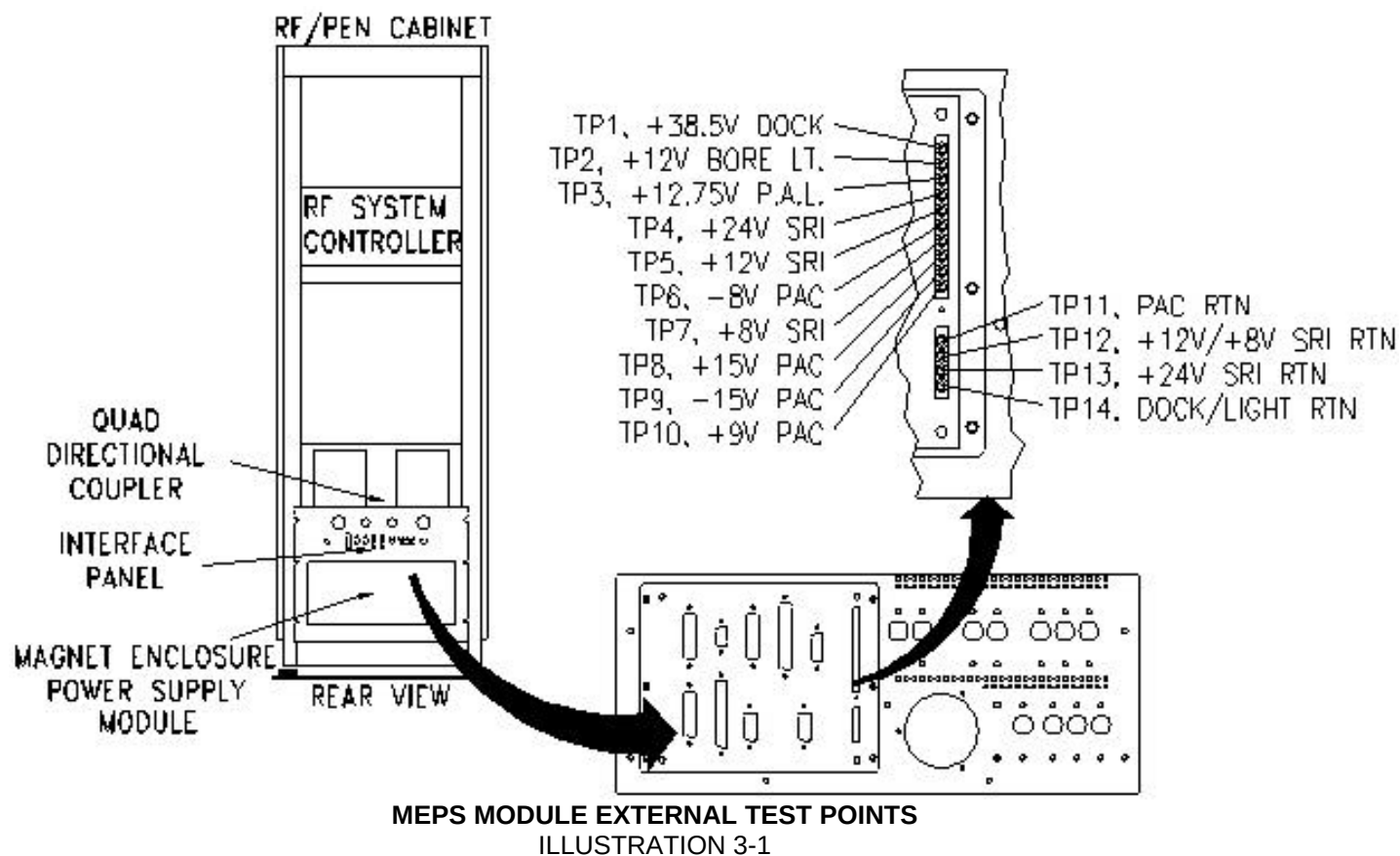
Digital Multimeter, Fluke 3030, or equivalent , 46-194427P49



Possible equipment damage. Damage may occur to the PAC board if power is connected to the low current power supply before this procedure has been completed.

3- HARDWARE SETUP FOR MEASUREMENT

1. Locate the back of the MEPS module and the test points there (see Illustration 3-1).



4- VOLTAGE MEASUREMENT

1. Connect DMM to appropriate test points on the back of the MEPS module.
2. Measure voltages per specifications in Table 4-1.

TABLE 4-1
PAC VOLTAGE ADJUSTMENTS

Power Supply	Probe Connection (on LCPS Board)	Voltage Specification	Adjustment	Ripple Spec. (Max P/P)
+9 Vdc	(+) TP5; (-) TP6	+9Vdc $\pm$ 0.1Vdc	R5	0.18V
-15 Vdc	(+) TP4; (-) TP2	-15Vdc to -16Vdc	R4	0.3V
+15 Vdc	(+) TP1; (-) TP2	+15Vdc to +16Vdc	R1	0.3V
-8 Vdc	(+) TP3; (-) TP2	-8Vdc $\pm$ 0.1Vdc	R2	0.16V

Note

For use with the PAC only, adjust the +9 Vdc power supply to +8.0 Vdc.

Note

For use with the Fiber Optic Repeater and PAC, adjust the +9 Vdc power supply to +9.0 Vdc.

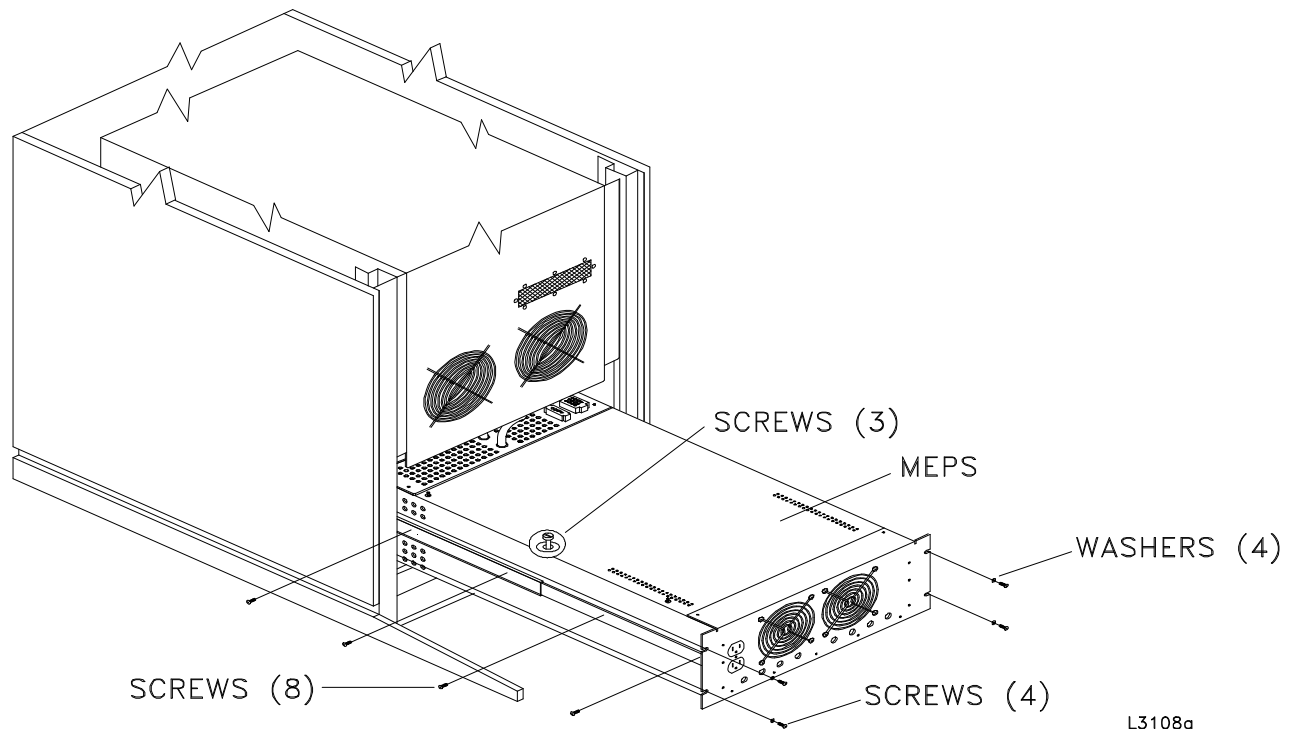
Note

All measurements taken at power supplies are done with the PAC assembly or Fiber Optic Repeater disconnected.

3. Measure ripple to ensure that it is within specification.
4. Remove DMM leads from the back of the MEPS Module.
5. If any of the power supplies need adjustment, continue with the procedure.

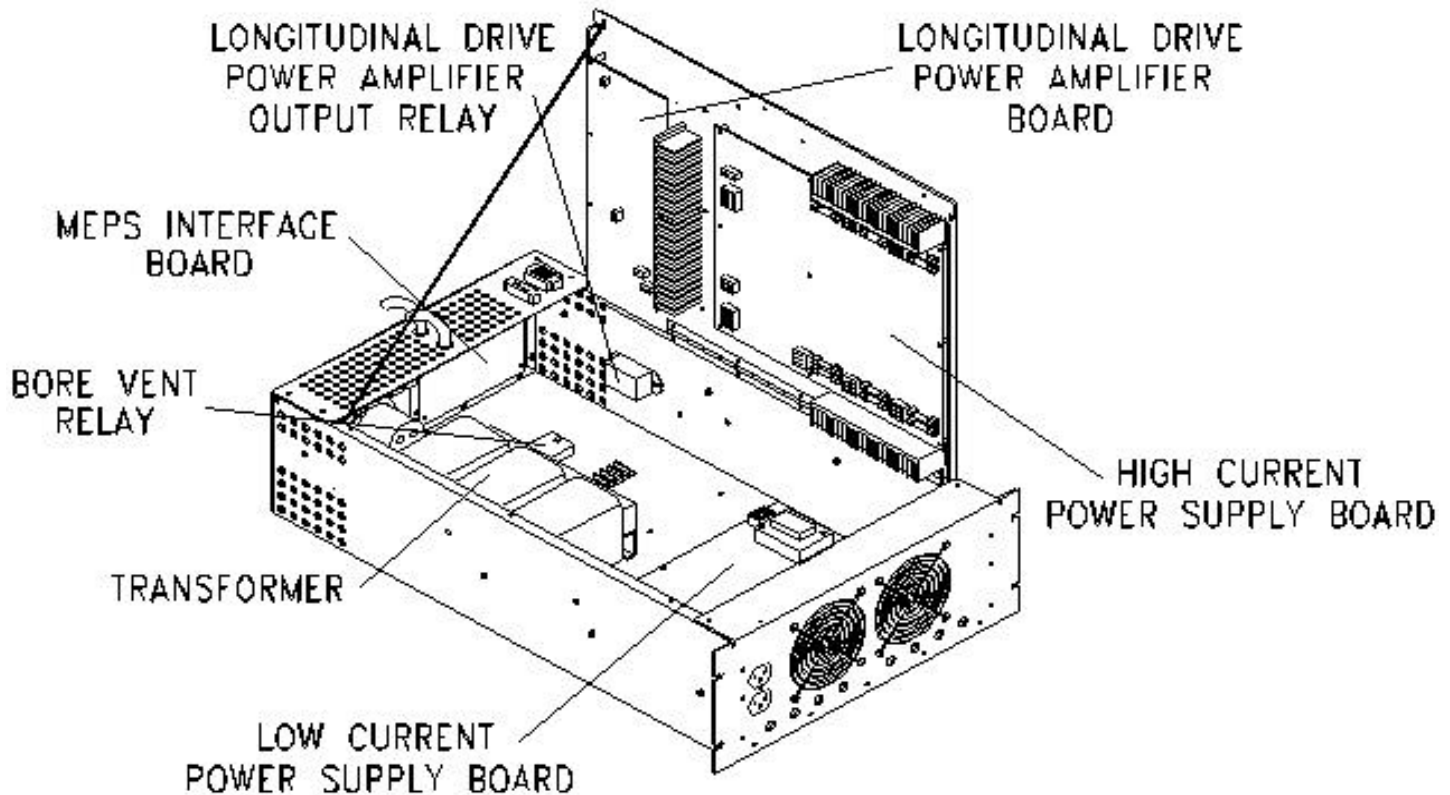
5- HARDWARE SETUP FOR ADJUSTMENTS

1. Remove the front door from the RF/PEN Cabinet.
2. Remove the four screws and four #10 flat washers securing the MEPS module to the cabinet.
3. Slide the MEPS module out the front of the cabinet (see Illustration 5-1).



MEPS MODULE IN RF/PEN CABINET
ILLUSTRATION 5-1

4. Loosen the three 4-40 captive screws on the top cover of the MEPS module and open the lid (see Illustration 5-1). Locate the appropriate board for the adjustments you need to make (see Illustration 5-2).



MAGNET ENCLOSURE POWER SUPPLY COMPONENTS
ILLUSTRATION 5-2

- For adjustments to the PAC and SRI Power Supplies, locate the Low Current Power Supply Board and proceed to Section 6- PAC and SRI Voltage Adjustments.
- For adjustments to the Patient Alignment Light, Bore Light, or Longitudinal/ Dock Power Supplies locate the High Current Power Supply Board and proceed to Section 7- Patient Alignment Light, Bore Light, and Longitudinal/ Dock Voltage Adjustments.

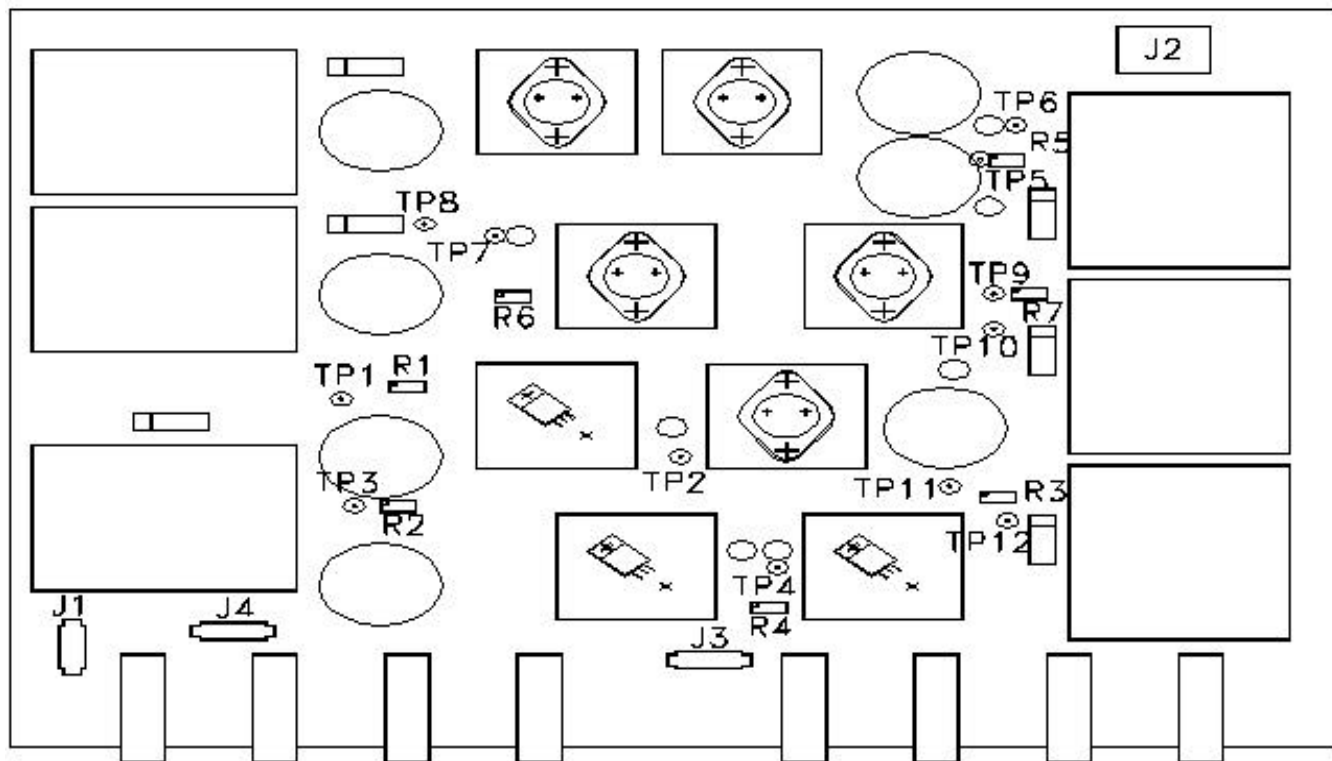
6- PAC AND SRI VOLTAGE ADJUSTMENTS

WARNING!

ELECTRIC SHOCK HAZARD!

POTENTIALLY LETHAL VOLTAGES MAY EXIST THE IMMEDIATE AREA.

1. Connect DMM to appropriate test points on the Low Current Power Supply Board located on the bottom of the MEPS. See Illustration 6-1 and Table 6-1 (for PAC) and Table 6-2 (for SRI).



LOW CURRENT POWER SUPPLY BOARD WITH PAC & SRI POWER SUPPLY TEST POINTS
ILLUSTRATION 6-1

TABLE 6-1
SRI VOLTAGE ADJUSTMENTS

Power Supply	Probe Connection (on LCPS Board)	Voltage Specification	Adjustment	Ripple Spec. (Max P/P)
+24 Vdc	(+) TP5; (-) TP6	+24V $\pm$ 0.1 V	R6	0.48V
+12 Vdc	(+) TP4; (-) TP2	+12V $\pm$ 0.1 V	R7	0.24V
+8 Vdc	(+) TP3; (-) TP2	+ 8V $\pm$ 0.1 V	R3	0.16V

TABLE 6-2
SRI VOLTAGE ADJUSTMENTS

Power Supply	Probe Connection (on LCPS Board)	Voltage Specification	Adjustment	Ripple Spec. (Max p-p)
+24 VDC	(+) TP5; (-) TP6	+24V $\pm$ 0.1 V	R6	0.48V
+12 VDC	(+) TP4; (-) TP2	+12V $\pm$ 0.1 V	R7	0.24V
+8 VDC	(+) TP3; (-) TP2	+ 8V $\pm$ 0.1 V	R3	0.16V

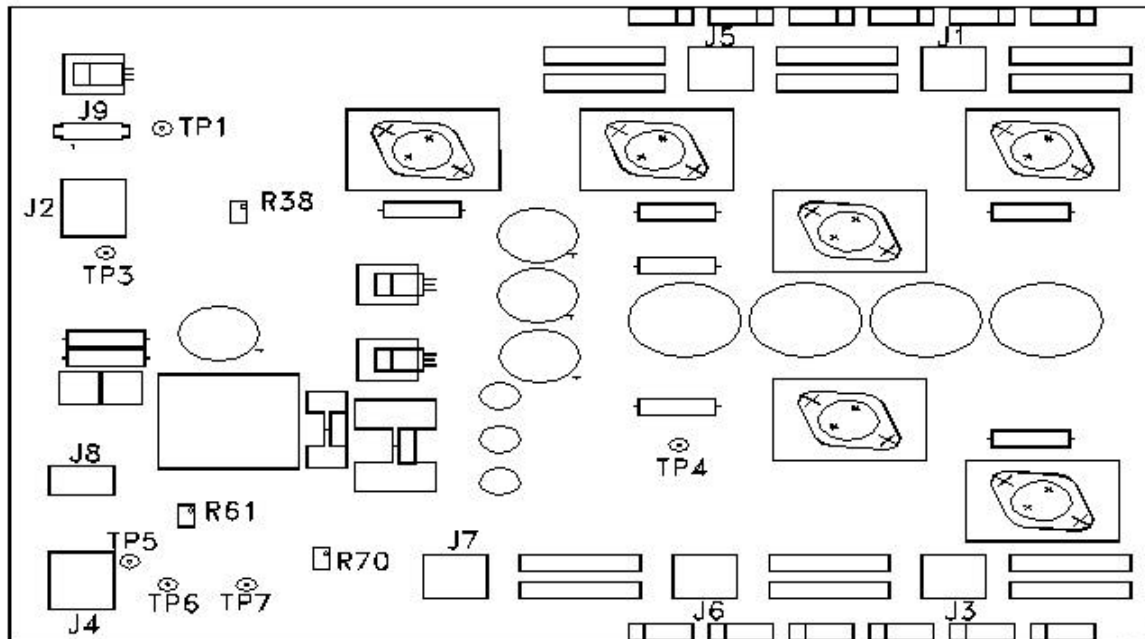
2. Adjust voltage per specification in Table 6-1 (for PAC) and Table 6-2 (for SRI).
3. Measure ripple to ensure that it is within specification.
4. Remove DMM leads from the board.
5. If any of the High Current Power Supply power supplies also need adjustment, continue with Section 7. Otherwise, go to Section 8.

7- PATIENT ALIGNMENT LIGHT, BORE LIGHT, AND LONGITUDINAL/DOCK VOLTAGE ADJUSTMENTS

WARNING!

ELECTRIC SHOCK HAZARD!
POTENTIALLY LETHAL VOLTAGES MAY EXIST THE IMMEDIATE AREA.

1. Connect DMM to appropriate test points on the High Current Power Supply Board located on the lid of the MEPS. See Illustration 7-1 and Table 7-1.



HIGH CURRENT POWER SUPPLY BOARD TEST POINTS

ILLUSTRATION 7-1

TABLE 7-1
HIGH CURRENT POWER SUPPLY VOLTAGE ADJUSTMENTS

Power Supply	Probe Connection (on HCPS Board)	Voltage Specification	Adjustment	Ripple Spec. (Max P/P)
+15V Patient Alignment Light	(+) TP5; (-) TP7	+12.75V to +16Vdc	R70	0.26V
+12V Bore Light	(+) TP6; (-) TP7	+12V to +15Vdc	R61	0.24V
+38.5V Longitudinal/ Dock	(+) TP3; (-) TP1	$\pm 38.5 \text{ V} \pm 0.1\text{Vdc}$	R38	0.77V

2. Adjust voltage per specification in Table 7-1.
3. Measure ripple to ensure that it is within specification.
4. Remove DMM leads from the board.
5. If any of the Low Current Power Supply power supplies also need adjustment, continue with Section 6- PAC and SRI Voltage Adjustments. Otherwise, proceed to Section 8.

8- SYSTEM RESTORATION

1. Close lid and tighten the three 4-40 captive screws on lid.
2. Slide the MEPS module back into the RF/PEN Cabinet and replace the four screws and four #10 flat washers, securing the module.
3. Install the RF/PEN Cabinet front door.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	July 22, 1998	M. Whitlow	Initial Conversion from Toolbook to Word.
1	January 27, 1999	K. Keshena	Updated styles.